

First Year Summary

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May 2026

1 Insights

1.1 Building tools, figuring out how things work, and conveying my insights to other people brings me immense joy and fulfillment.

I discovered that I enjoy building tools (mathematical/mental models etc.) to help people better understand things and solve problems. The most fulfilling moments I had was when I came up with useful analogies to convey difficult ideas in an intuitive way: the drawer analogy to explain how pointers work, the interference machine analogy to explain the mathematical structure behind Simon's algorithm, the Christoffel triangle method to quickly compute the Christoffel symbols in terms of the metric components, the moving clock story to explain why we cannot talk about time without talking about space etc.

I also really enjoy figuring out how things work. For example, when I finally came up with a framework to explain why high-frequency voltage led to a consistently higher-than-expected current, and when I finally found the reason behind why our torsional-oscillator model had a quadratic shaped residual, I felt tremendously satisfied and at the same time tremendously surprised that, when I think about a problem long enough, I too, could independently come up with a solution.

1.2 Understanding & Mastery Takes Tremendous Hard Work and Effort.

Understanding something is like making your way through a dense forest. If you want to make it through the forestry you have to slash the obstacles ahead and take a few wrong turns before the view becomes clear (for there to be an aha moment). In addition, finding your way out once doesn't mean that you will automatically make it out the second time. It takes multiple trips before you internalize the landscape and know it as well as the back of your hand. Hence there is no understanding and mastery that does not take a tremendous amount of hard work and effort.

Understanding this helped me build my inner scorecard and let go of test anxiety. Tests are just reflections of how well I master something, and mastery is a marathon. If I put in decades of effort, I believe that I will eventually master something like I know the back of my hand.

1.3 It is Extremely Important to Document and Ponder Your Questions, Observations, and Ideas

Deep insights come from thinking about problems for a very long time.

It is never difficult for me to come up with questions, but previously, I did not have the habit of recording them, let alone think about them for a long time. However, during this year, I began recording my questions in a question book and frequently visited them. I can't stress how valuable this habit is because my question book provided me so much intuition and insights that couldn't be gained elsewhere.